Ray Reyes
Kiet Tran
Thanh Nguyen
Professor Roth

Humans Are Bad For The Air Quality

Our thesis explores the idea that human activity harms the environment, especially the air quality. We support this by analyzing data that demonstrates a correlation between population growth and PM2.5 levels. Additionally, we examine how increased energy consumption contributes to higher PM2.5 levels. Finally, we highlight the positive correlation between Energy Consumption Per Capita (EPC) and PM2.5 Per Capita (PPC), illustrating that greater individual energy use directly contributes to elevated PM2.5 levels.

To convince the reader of our thesis that human activity negatively impacts the environment, we will present a structured argument supported by data and analysis. First, we will explore the relationship between population growth and PM2.5 levels. We did this by looking at the density level of each county and the PM 2.5 levels of those counties and then finding the correlation level. Next, we will examine the impact of energy consumption on air quality, using PM2.5 levels as a key indicator to show how rising energy use, particularly from natural gas, contributes to worsening air pollution. Again, we will use the residential gas consumption level per county and then compare it with the PM 2.5 levels. We will then establish the positive correlation between Energy Consumption Per Capita (EPC) and PM2.5 Per Capita (PPC), demonstrating that higher individual energy use results in greater pollution per person. We first calculate the EPC and PPC for each county and then show the relationship between the two parameters.

First, we look at the density level of each county in the US. With the data from Simple Maps, we could identify each county's most up-to-date density level by their county FIPS. Next, we gathered the air quality data from the EPA and extracted the PM2.5 24-hour (PM2.5 level average over 24 hours). We sorted the data by increasing density and then log-scale the data. We then did a linear regression model over the 2 data and produced Figure 1: Relationship Between Density and PM2.5. From this figure, we can identify a positive relationship between population density and PM 2.5, suggesting that population density has a negative influence on air quality.

For our second figure, we wanted to identify the relationship between energy consumption and pollution. Similarly to our previous figure, we measure pollution by examining particulate matter emissions(PM 2.5). The measurement of energy consumption looks at how much natural gas is consumed by residents and is measured in TcF units. We use data from the EPA for our particulate matter data and the Department of Energy for natural gas consumption. The second figure shows a positive relationship between the two variables. This is because as people consume more and more energy there is an ever-increasing amount of pollution that comes with that energy usage. The blue dots represent the countries in our data set and we chose not to label them as it

would have overwhelmed the graph and made it illegible. We added a trendline to make the relationship between energy consumption and pollution.

For the third argument, we look at per capita energy usage and pollution. Similar to the last graph we continue using residential natural gas consumption to represent energy consumption and use particulate matter to measure pollution. The difference in this graph is that we divide these variables by the population in that county. It produces two new variables that we compare, energy consumption per capita and particulate matter emissions per capita. These new variables allow us to estimate this relationship at an individual level. Each blue dot represents the average per country per person and with the help of the trendline we can see the positive relationship between the two.

One counter-argument could be that technological advancements and innovations, such as cleaner energy sources and improved efficiency in energy use lower the environmental impact of population growth and energy consumption. While this is valid, our analysis primarily reflects current usage patterns of natural gas and does not account for future technological shifts. Another potential limitation is that PM2.5 levels may also be influenced by factors beyond population density and residential energy consumption, such as industrial activities, transportation, and natural phenomena like wildfires or dust storms. Addressing these gaps would require incorporating additional variables and data sources into our analysis to isolate the impact of each factor. Expanding our study to include other energy types and their respective emissions could strengthen our conclusions.

Our data indicates that an increasing population is directly linked to greater harm to the environment. As the number of people grows, so does the demand for resources, energy, and infrastructure, all of which contribute to pollution and environmental degradation. To address this, our research highlights the need for stricter air quality regulations aimed at reducing the pollution we emit. While population growth is inevitable, its impact can be managed if we collectively adopt more sustainable practices. Simple actions, such as being mindful of our energy consumption and reducing waste, can contribute significantly to this effort.

References

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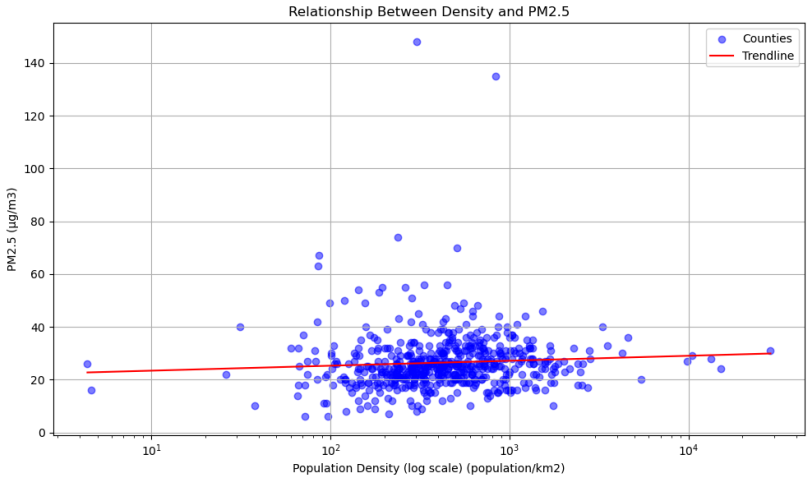

Figure 1: Relationship Between Population Density (population/km2) and PM2.5 (µg/m3)

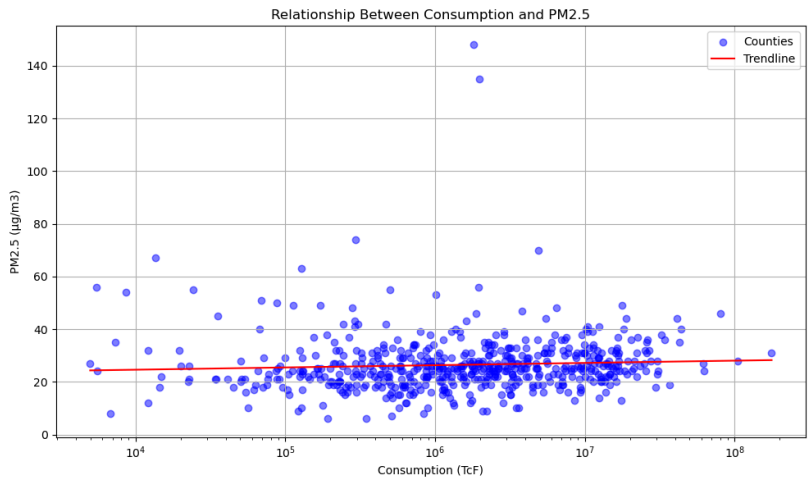

Figure 2: Relationship between residential gas consumption (TcF) and PM2.5 (µg/m3)

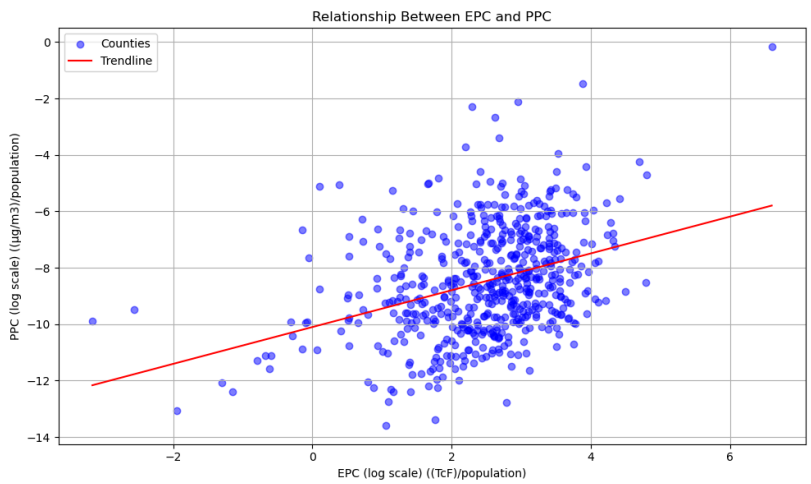

Figure 3: Relationship between EPC (energy consumption per capita) ((TcF)/population) and PPC (pollution per capita) ((µg/m3)/population)

Team Member Contributions: All three members looked for the data, Kiet Tran and Thanh Nguyen primarily worked on the code and Ray Reyes primarily worked on the report and presentation.